Bayesian weak form Estimation of Nonlinear Dynamics (BENDy)

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References

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Contributions

- Extended prior work [1] to nonlinear-in-parameters systems.
- Formulated analytic expressions for an approximate likelihood and its derivatives.
- Implemented efficient and application friendly Julia [2] code.
- Accommodated more general noise distributions.
- Extended to a Bayesian framework with automatic objective priors.
- Guaranteed consistency of the BENDy estimator.

Weak Form Parameter Estimation

Goal: Estimate the unknown parameters $\mathbf{p} \in \mathbb{R}^J$.
Given a (possibly *nonlinear*) system of differential equations

$$\dot{\boldsymbol{u}} = \boldsymbol{f}(\mathbf{p}, \boldsymbol{u}, t) \tag{1}$$

where $\boldsymbol{f} : \mathbb{R}^J \times \mathbb{R}^D \times \mathbb{R} \to \mathbb{R}^D$, $\boldsymbol{u} \in \mathbb{R}^D$ and $t \in [0, T]$.
Transform eq. (1) to its weak form by taking the inner product of both sides with a test function φ along each dimension

$$-\langle \dot{\varphi}, \boldsymbol{u} \rangle = \langle \varphi, \boldsymbol{f}(\mathbf{p}, \boldsymbol{u}, t) \rangle \tag{2}$$

Assume access to $M+1$ observations $\{(t_m, \mathbf{U}_m)\}_{m=0}^M$ where the data has been corrupted by noise $\varepsilon_m \overset{iid}{\sim} \mathcal{N}(\mathbf{0}, \Sigma)$.

Distribution	Data Description
Additive Gaussian	$\mathbf{U}_m = \mathbf{u}_m + \varepsilon_m$
Multiplicative LogNormal	$\mathbf{U}_m = \mathbf{u}_m \exp(\varepsilon_m)$

Discritization

Selecting specific test functions $\{\varphi\}_{k=1}^K$, the inner products can be approximated through quadrature.

$$-\dot{\mathbf{\Phi}}\mathbf{Q}\mathbf{U} \approx \mathbf{\Phi}\mathbf{Q}\boldsymbol{F}(\mathbf{U}, \mathbf{t}, \mathbf{p}). \tag{3}$$

where the following are defined for convenience:

Test Function Matrix	$\mathbf{\Phi} \in \mathbb{R}^{K \times (M+1)}$
Quadrature Matrix	$\mathbf{Q} \in \mathbb{R}^{(M+1)\times(M+1)}$
Observation Matrix	$\mathbf{U} \in \mathbb{R}^{(M+1)\times D}$
Matrix-valued Function	$\boldsymbol{F} : \mathbb{R}^{(M+1)\times D} \times \mathbb{R}^{M+1} \times \mathbb{R}^J \to \mathbb{R}^{(M+1)\times D}$

Approximate Distribution

In expectation, the weak residual should be small

$$\mathbf{0} \approx \mathbf{r}(\mathbf{p}) = \text{vec}\left[\dot{\mathbf{\Phi}}\mathbf{Q}\boldsymbol{F}(\mathbf{U}, \mathbf{t}, \mathbf{p}) + \mathbf{\Phi}\mathbf{Q}\mathbf{U}\right] \in \mathbb{R}^{KD} \tag{4}$$

when $\mathbf{p} \approx \mathbf{p}^*$. Notice, $\mathbf{r}(\mathbf{p}^*) \neq \mathbf{0}$ because:

- Inner products are *approximated* through quadrature.
- Data $\mathbf{U}$ is noisy.

Through a *linearization* with respect to $\boldsymbol{u}$ about the data $\mathbf{U}$, one can obtain the approximate distribution of the weak residual

$$\mathbf{S}(\mathbf{p})^{-1/2}\mathbf{r}(\mathbf{p}) \overset{approx}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{5}$$

where an analytic expression for $\mathbf{S}$ follows from the linearization and the noise distribution, see [?].

Maximum a Posteriori Estimator

Often a prior for the parameters $\mathbf{p} \sim \Pi$ is known. Alternatively, one can utilize an *objective prior* [3–5].
The approximate negative log-posterior is proportional to

$$\ell(\mathbf{p}) \propto \text{logdet}(\mathbf{S}(\mathbf{p})) + \mathbf{r}(\mathbf{p})^\top \mathbf{S}(\mathbf{p})^{-1}\mathbf{r}(\mathbf{p}) + \pi(\mathbf{p}). \tag{6}$$

The unknown parameters can be obtained via optimization

$$\hat{\mathbf{p}}_{\text{MAP}} = \underset{\mathbf{p}}{\text{argmin}}\Big(\ell(\mathbf{p})\Big). \tag{7}$$

Consistency

Recall, the *Gaussian approximation* is obtained via a *linearization* with respect to $\boldsymbol{u}$; this leads to bias, even when data is dense

$$\lim_{M\to\infty} \mathbb{E}[\hat{\mathbf{p}}_{\text{MAP}}] \neq \mathbf{p}^*. \tag{8}$$

To address this, correct by using more terms in the expansion

$$\mathbf{c}(\mathbf{p}) = (\mathbf{\Phi}\mathbf{Q}\nabla_p \boldsymbol{F}(\mathbf{t}, \mathbf{U}, \mathbf{p}))^\dagger \,\text{vec}\left[\sum_{\alpha=2}^{\alpha_{\text{trunc}}} \frac{1}{\alpha!}\mathbf{\Phi}\mathbf{Q}\mathbb{E}\left[\nabla_{\boldsymbol{u}}^\alpha \boldsymbol{F}(\mathbf{t}, \mathbf{U}, \mathbf{p}) \times_1 \mathcal{E} \cdots \times_\alpha \mathcal{E}\right]\right],$$
$$\hat{\mathbf{p}}_{\text{C-MAP}} = \hat{\mathbf{p}}_{\text{MAP}} - \mathbf{c}(\hat{\mathbf{p}}_{\text{MAP}}). \tag{9}$$

Equation (9) can be approximated via analytic expressions for $\nabla_{\boldsymbol{u}}^\alpha \boldsymbol{F}$ and known centered moments for Gaussians or by sampling.
In a neighborhood of $\mathbf{p}^*$, *consistency* is obtained

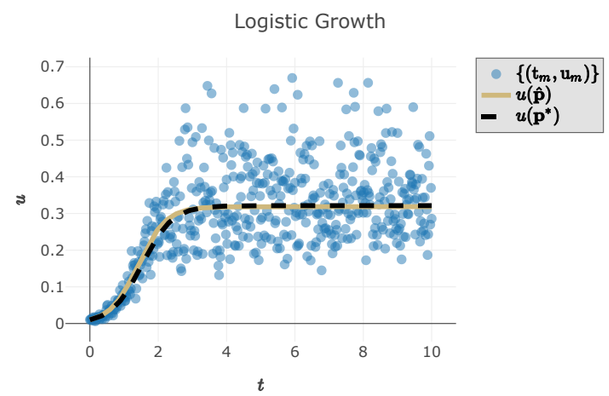
$$\lim_{M\to\infty} \mathbb{E}[\hat{\mathbf{p}}_{\text{C-MAP}}] = \mathbf{p}^*. \tag{10}$$

Example: Logistic Growth

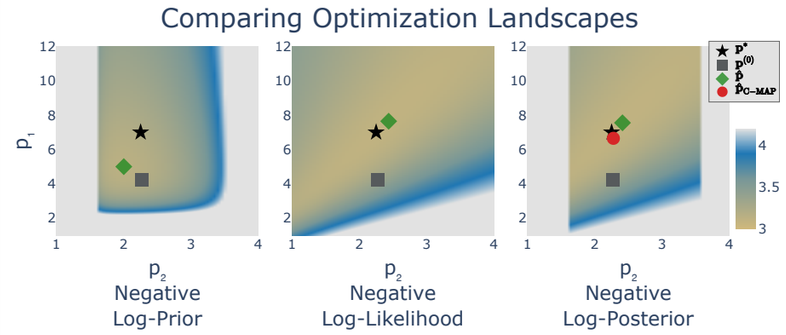
Consider the ordinary differential equation

$$\dot{u} = p_1 u - p_2 u^2 \tag{11}$$

where $t \in [0, 10]$ and $\mathbf{p}^* = [2.25, 7.0]^\top$. Data is observed on a uniform grid $\{(m\Delta t, \mathrm{u}_m)\}_{m=0}^M$ corrupted by multiplicative LogNormal noise.



Given the priors $p_1 \sim 2(\beta(2,5) + 0.8)$ and $p_2 \sim \text{Fréchet}(\alpha = 3, \theta = 5.503)$, one can now estimate $\mathbf{p}$ with eq. (7) and eq. (9).



Optimizing the prior or the likelihood alone leads to an estimate that is less accurate than either the MAP or its correction.

Objective		Relative Parameter Error
Prior	$\hat{\mathbf{p}}_\Pi$	27.4%
Likelihood	$\hat{\mathbf{p}}_{\text{MLE}}$	9.1%
Posterior	$\hat{\mathbf{p}}_{\text{MAP}}$	7.9%
Corrected	$\hat{\mathbf{p}}_{\text{C-MAP}}$	**4.7%**
OE-LS	$\hat{\mathbf{p}}_{\text{OE-LS}}$	7.3%